

## EXPERIMENT 11: IMAGE FUSION

AIM:

To perform image fusion using python in google colab.

MATERIALS REQUIRED:

Google colab software (python)

CODE:

### 1)Python Code (Weighted Fusion)

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

# Load images

img1 = cv2.imread(' bone fracture 1.jpg')

img2 = cv2.imread(' bone fracture 2.jpg')

# Resize images to same size

img2 = cv2.resize(img2, (img1.shape[1], img1.shape[0]))

# Image fusion

alpha = 0.6

fused = cv2.addWeighted(img1, alpha, img2, 1-alpha, 0)

# Display images

plt.figure(figsize=(10,4))

plt.subplot(131)

plt.title("Image 1")

plt.imshow(cv2.cvtColor(img1, cv2.COLOR_BGR2RGB))

plt.axis('off')

plt.subplot(132)

plt.title("Image 2")

plt.imshow(cv2.cvtColor(img2, cv2.COLOR_BGR2RGB))

plt.axis('off')

plt.subplot(133)

plt.title("Fused Image")
```

```
plt.imshow(cv2.cvtColor(fused, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```

**OUTPUT:**



## 2)Python Code (Wavelet Fusion)

```
import cv2
import numpy as np
import pywt
import matplotlib.pyplot as plt

# Load images
img1 = cv2.imread('bone fracture 1.jpg', 0)
img2 = cv2.imread('bone fracture 2.jpg', 0)

# Resize images
img2 = cv2.resize(img2, (img1.shape[1], img1.shape[0]))

# Wavelet decomposition
coeffs1 = pywt.dwt2(img1, 'haar')
coeffs2 = pywt.dwt2(img2, 'haar')
LL1, (LH1, HL1, HH1) = coeffs1
LL2, (LH2, HL2, HH2) = coeffs2

# Fusion rule (maximum selection)
LL = (LL1 + LL2) / 2
LH = np.maximum(LH1, LH2)
HL = np.maximum(HL1, HL2)
HH = np.maximum(HH1, HH2)
```

```

# Reconstruction
fused = pywt.idwt2((LL, (LH, HL, HH)), 'haar')

# Display results
plt.figure(figsize=(8,4))
plt.subplot(131)
plt.title("Image 1")
plt.imshow(img1, cmap='gray')
plt.axis('off')
plt.subplot(132)
plt.title("Image 2")
plt.imshow(img2, cmap='gray')
plt.axis('off')
plt.subplot(133)
plt.title("Fused Image")
plt.imshow(fused, cmap='gray')
plt.axis('off')
plt.show()

```

**OUTPUT:**



### 3)Python Code (PCA Fusion)

```

import cv2
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
img1 = cv2.imread(' bone fracture 1.jpg', 0)

```

```
img2 = cv2.imread(' bone fracture 2.jpg', 0)
img2 = cv2.resize(img2, (img1.shape[1], img1.shape[0]))
# Stack images
stacked = np.stack((img1.flatten(), img2.flatten()), axis=1)
# PCA
pca = PCA(n_components=1)
fused = pca.fit_transform(stacked)
fused_image = fused.reshape(img1.shape)
plt.imshow(fused_image, cmap='gray')
plt.title("PCA Fused Image")
plt.axis('off')
plt.show()
```

**OUTPUT:**



**RESULT:**

Thus the image fusion were performed by google colab.